CS486 – 20CTT1 – Lab Final

75 minutes

Notes

- Put ALL SQL statements to ONE text (SQL) document named as W10_[StudentID]_yy.sql (yy is a self-grading mark 01-10)
- Comment Student ID and Full name on top of file
- Run ALL lines in your script without any error
- USE the CORRECT database
- Note that **C** is a constraint, **Q** is a query, **F** is a function, **S** is a stored procedure.

Database

Database name: LabFinal_[StudentID]_[FullName]
Tables:

- **Wallets:**
  - ID: int, auto increment (*), primary
  - Address: nvarchar(255), not null, unique
  - DateCreated: date, not null, must be in format YYYY-MM-DD
  - NetworkID: nvarchar(3), not null, foreign key reference to NetworkID in Networks table
  - Balance: float, not null, default = 0, >=0 **(C1)** (the current balance of the account in its network's currency)

ID	Address	DateCreated	NetworkID	Balance
1	0xf3298c3d8aecb82acd77e9218d3dd6	2014-07-29	BSC	59.492397
2	0x9166d66f4057107fec18acfffd9e73cb	2016-10-22	ETH	1.702557
3	0x4335A6DCcb02F7Ab429aeDC01B802	2020-11-05	ETH	2.026917
4	0x3487a7e19345eabe99a491c08abf86c	2018-03-07	TRC	289855.072464
5	0xf55e931365a1b7b696fbad08dfc817e	2018-03-26	BSC	300.00
6	0xd204270a8124c0d4d621dacbb6079e	2022-01-22	SBC	10
7	0xafb8c11849c848a6ebd000486a41231	2017-02-11	POL	100000000

- **Networks:**
  - NetworkName: nvarchar(50), not null
  - NetworkID: nvarchar(3), not null, primary
  - Currency: char(5), not null
  - UsdRatio: float, not null (Note: the USD ratio means that a single token in this currency is equivalent to the amount of USD).

NetworkName	NetworkID	Currency	UsdRatio
Ethereum Mainnet	ETH	ETH	1480.08
Binance Smart Chain	BSC	BNB	278.17
Polygon Mainnet	POL	MATIC	0.8852
Smart Bitcoin Cash	SBC	BCH	156.09
Tronix	TRC	TRX	0.069

- **Transactions**:
  - ID: int, auto increment (*), primary
  - TransTime: DateTime, not null, with format YYYY-MM-DD HH:MM:SS
  - FromWalletID: int, not null, foreign key reference to wallet's ID in Wallet table
  - ToWalletID: int, not null, foreign key reference to wallet's ID in Wallet table
    - Note that if FromWalletID and ToWalletID are the same, it means that the wallet is trying to convert the currency
  - SendCurrency: char(3), not null
  - ReceiveCurrency: char(3), not null
  - SendAmount: int, not null, > 0 **(C2)** (Note: the sending amount is in the sending currency)

ID	TransTime	FromWalletID	ToWalletID	Send	Receive	SendAmount
1	2021-10-10 05:35:31	1	1	USD	BNB	100000
2	2022-04-15 12:18:20	2	2	USD	ETH	4000
3	2022-05-16 23:11:33	3	3	USD	ETH	3000
4	2022-06-11 07:30:39	4	4	USD	TRX	20000
5	2022-07-23 10:06:11	1	5	BNB	BNB	300
6	2022-07-29 02:17:55	2	6	ETH	BCH	1

(*) Auto increment: IDENTITY(1,1)

Constraints

- **C3:** TransTime of a transaction must occur after the FromWalletID and ToWalletID were created.
- **C4:** The FromWalletID must have enough balance to transfer the SendAmount in a transaction.
- **C5:** The send currency in a transaction must be 'USD' or match the network currency of the FromWalletID

Requirements:

1. Create the database (0.25)
2. Create tables with primary keys, foreign keys, and unique index (0.5)
3. Insert sample data (use the provided script)
4. Implement **C1, C2**, **C3, C4, C5** constraints (0.25, 0.25, 0.5, 0.5, 0.5)
5. **Q1**: Find the network's names that have no transaction (1.0)
   - Polygon Mainnet
6. **Q2**: Which currency (except 'USD') has the largest total amount (in terms of USD) in all transactions (1.0)
   - BNB
7. **F1:** Check if a network is not in use (no transaction using its currency)? (1.0)
   - Polygon Mainnet is not in use.
8. **F2**: Check if a wallet (ID) is a **fraud** (wallet that has a balance different from accumulating transactions) (1.0)
   - Wallet #6 is a fraud since its balance should be 9.482222 instead of 10.
   - Wallet #7 is a fraud since its balance should be 0 instead of 100000000.
9. **S1**: Create a **birthday gift** for a wallet by transferring 100 MATIC from the wallet with ID #7 (2.0)

- Validate constraints and check if today is the wallet’s birthday.
- Increase the wallet’s balance by 100 MATIC (remember to convert to the correct currency).
- Decrease wallet #7 ‘s balance by 100 MATIC.
- Create transaction record
- Resolve concurrency access

10. Put begin, rollback and commit transaction statements to appropriate positions in **S1**. Which isolation level can cause the deadlock when S1 is executed two times at the same time? Simulate the deadlock by using the statement waitfor delay ’00:00:05’ at a suitable position in S1. (1.0)
11. Drop the database (0.25)